Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/18/2026
PI Name:	Stankovic, Aleksandra Ph.D.		
Project Title:	Modeling Individual and Team Problem-Solving Skills for the Establishment of Training Metrics, Standards, and Guidelines for Long-Duration Space Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC018N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	03/09/2020	**End Date:**	08/30/2025
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Assignment:	NOTE: End date changed to 08/30/2025 per NSSC information (Ed., 12/30/24) NOTE: End date changed to 03/08/2025 per JSC Grant-Technical Officer (Ed., 2/26/24) NOTE: End date changed to 03/08/2024 per A. Beitman/JSC (Ed., 2/6/23)		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI), Drs. Strangman and Robinson were removed from the list of Co-Investigators in this reporting period (Ed., 3/19/26).		
COI Name (Institution):	Caldwell, Barrett (Purdue University)		
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Task Description:	The goal of this project was to investigate how teamwork, communication, and collaborative problem-solving support operational performance and medical autonomy in isolated, confined, and extreme (ICE) environments relevant to long-duration human spaceflight. This research examined how teams coordinate under conditions of uncertainty, time pressure, and environmental stress, with a particular focus on medical care delivery, construction tasks, science return, and emergency response scenarios in remote and resource limited settings. The project sought to identify behavioral, organizational, and technological factors that promote effective collaboration, resilience, and shared situational awareness in spaceflight-relevant contexts. This project conducted qualitative and mixed-methods investigations across multiple terrestrial analog settings with several teams, including in hospital and wilderness medicine contexts, in complex construction and repair scenarios in the oil and gas industry, in field-based scientific return operations in the remote Himalyas and at the South Pole research station, and in emergency response situations in extreme environments such as Antarctica. These settings provided ecologically valid insights into how teams adapt to constrained resources, limited expertise, and delayed communication. Findings demonstrated that successful team performance depends on clear role definition, adaptive leadership, shared mental models, and effective use of decision-support tools. Across all environments and operational contexts, teams benefited from structured communication strategies and flexible automation support that enhanced, rather than replaced, human judgment. The project also advanced conceptual and practical frameworks for integrating automation into exploration medicine. These frameworks emphasize graduated autonomy, transparency, and alignment with team workflows, informing the design of future medical decision-support systems for spaceflight. Additional outputs of this work included a novel simulation and training capability leveraging a digital twin environment for multi-team coordination in Lunar and Cis-Lunar operations. It is our hope that, collectively, findings from this project will help inform future functional requirements, training and support strategies, and team-based concepts of operation to enable safer and more effective long duration human space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are undertaking an operationally-relevant assessment of team problem-solving skills in the context of upcoming Cis-Lunar exploration, with the ultimate goal of contributing towards the establishment of future autonomous team-based concepts of operations and functional design requirements for system development, operations, and training.
	Summary: This project conducted qualitative and mixed-methods investigations across multiple terrestrial settings that offered high-fidelity operational analogs of space exploration, including low-resource medicine contexts, complex construction and repair scenarios, emergency response, and scientific field investigations in remote environments. Deploying our research in these domains provided ecologically valid insights into how teams adapt to constrained resources, isolation/confinement, and delayed communication. Methods: Using relevant Earth-bound analogs for anticipated Lunar/Cis-Lunar activities across 4 key anticipated activity domains (construction/repair, science return, medical operations, and emergency response), this project aimed to (1) determine how problem-solving skills are developed and used across a variety of operational tasks and situations relevant to future long-duration spaceflight exploration, and (2) identify which team and human-autonomous systems collaboration skills are most important to performing tasks in both normal and emergency conditions. The project included three research phases: (1) literature review and translational methodology development; (2) in-situ data collection (e.g., contextual inquiry, observation, interviews, surveys) and data analysis; and (3) development of models, concepts of operations, and recommendations for effective teaming in Lunar and Cis-Lunar operations. The first phase of this investigation involved an extensive literature review that resulted in the identification of processes and outcomes of interest, along with key knowledge gaps and needs (Harris et al., 2024). This initial definition phase guided our methodology development, and the refinement of our data collection tools (e.g., questionnaires, interview guides), which included the incorporation of surveys developed and validated as part of the BHP Exploration Measures battery (i.e., team performance, team cohesion, psychological safety, and team processes). In phase two, we employed a mixed-methods, translational research approach to study how operational teams in extreme environments work together to solve complex, real-time problems through a combination of semi-structured interviews, embedded task observation, and surveys. Teams were followed for a period of 4-8 weeks, depending on the length of their team activities and deployments. Team members completed weekly survey measures assessing team performance, cohesion, psychological safety, and teamwork processes using the BHP-developed questionnaires. Questionnaire data were aggregated upon analysis. Participants also completed a 1.5 hr semi-structured interview to discuss in detail their experiences with team-based problem-solving, including skills and processes, along a variety of operational tasks and situations relevant to future long-duration spaceflight exploration. These interviews focused on the full continuum of the team decision-making process from problem identification to group-based solution ideation and consensus building, to plan execution, outcome evaluation, and replanning. These methods were adopted because they provided higher fidelity examinations of operational conditions than laboratory testing, and could be highly customizable to different situations and contexts. The goal of this data collection was to characterize the relationships among team members, and examine how factors such as tasks, roles, operational situations, work phases, experience, training, collaboration with autonomous tools, and other key variables identified through the literature analysis phase relate to the formation and appraisal of various team-based skills and needs. To capture team decision-making across a range of anticipated future lunar exploration activities, we recruited teams from several terrestrial analog settings representing anticipated Cis-Lunar operational functions, including emergency response, construction and repair, science return, and multi-role long-duration operations. The first cohort was comprised of winter-over crewmembers at the Antarctic Amundsen-Scott South Pole Station. We enrolled three teams, each composed of three or more members: the fire response team (5 team members, followed for 9 weeks), the equipment and supply team (5 team members, followed for 9 weeks), and the science team (3 team members, including 1 remotely participating teammate, followed for 8 weeks); of the 10 total participants in this cohort, 5 were members of multiple teams, which was considered reflective of anticipated crew dynamics in future Moon missions, where astronauts are anticipated to preform multiple functions over the course of their missions. Additional cohorts included a four-person glaciology science team followed over a six-week Himalayan expedition (4 team members) and a construction team (5 participants) followed during a two-week oil-rig rotation (completing every-other-day assessments

Task Progress:

on account of the shortened deployment span).

Additionally, a doctoral research project was completed to examine how automation can be effectively integrated, specifically into medical teamwork scenarios relevant to long-duration spaceflight, where crews must provide complex care with limited personnel, equipment, and ground support. The project was motivated by the need to transition from Earth-dependent medical models to self-sufficient, crew-autonomous systems capable of supporting diagnosis and treatment in isolated, confined, and extreme environments. To address this challenge, the work employed a qualitative, human-systems integration approach that examined terrestrial care settings operating under constrained conditions. Observations and interviews were conducted across a range of medical environments (e.g., high- and low-resource medical settings including emergency medicine in hospital, wilderness, and rural clinics) to identify how teams coordinate under uncertainty, allocate authority, and incorporate decision-support tools into clinical workflows. Findings were synthesized into an automation framework for exploration medicine that emphasized graduated autonomy, transparency, and alignment with team processes, and that generated contextually grounded design requirements for future spaceflight medical automation systems.

Also, a software-based visualization and modeling capability was developed leveraging a digital twin framework of a representative Cis-Lunar mission concept involving multiple, distributed teams (e.g., ground-based MCC, Cis-Lunar orbital station, Lunar surface). The purpose of this tool is to provide a high-fidelity simulation platform to support operational concept definition, mission planning and real-time re-planning, and crew training, all synchronized across multiple agents. This modeling effort translated insights from terrestrial analog teams into a computer-based digital-twin simulation platform designed to model distributed team coordination and decision-making. Using systems engineering approaches and empirical data from field studies, the modeling environment simulated how teams allocate tasks, adapt roles, and respond to dynamic operational conditions. The resulting tool enables both pre-mission training and real-time mission planning by allowing teams and planners to explore alternative task distributions and decision pathways. This modeling capability provides a foundation for just-in-time training, adaptive mission replanning, and the development of task-specific training flows for exploration missions, thereby extending the empirical findings of the teamwork studies into operationally relevant simulation and decision-support tools.

Results and Discussion: Across sites and operational contexts, this study identified consistent patterns in how effective teams recognized problems, made decisions, planned, executed tasks, and learned from outcomes. High-performing teams demonstrated clear local authority structures combined with collaborative consultation, shared mental models of task priorities, and strong psychological safety that encouraged open communication of concerns. Teams were most resilient when they used simple, shared frameworks to assess problem criticality and when leadership roles flexibly shifted based on expertise. Conversely, breakdowns tended to occur when authority was ambiguous, procedures were outdated or poorly adapted to the environment, or interpersonal strain discouraged reporting of problems. Communication with remote experts also posed challenges when those experts were perceived as evaluative rather than supportive, which reduced trust and cooperation.

The project produced a set of cross-cutting concepts of operations and measurable indicators for effective team-based problem solving in exploration contexts. Key recommendations included a “consensus-default, expertise-override” decision model, embedded human-factors checks for fatigue and mood, distributed role adaptability, and short, structured debriefs to capture lessons learned. Effective teams demonstrated rapid problem recognition, broad participation in information sharing, flexible decision structures, and systematic knowledge capture. The findings also emphasized the importance of simple, resilient tools, redundant systems, and communication architectures that support asynchronous coordination under latency constraints. These results provide empirically grounded requirements for team training, operational procedures, and decision-support technologies for lunar and Mars missions, and establish measurable metrics that can be used to assess and train crew problem-solving performance.

For the medical automation doctoral research project, findings demonstrated that effective automation in spaceflight medical contexts must be designed as a collaborative teammate rather than as a replacement for human judgment. Across observed care environments, clinicians strongly preferred automation tools which supported role clarity, facilitated information sharing, and enhanced situational awareness without undermining human authority or trust. The resulting Automation Framework for Exploration Medicine provided evidence-based design guidelines for integrating automation into low-resource diagnostic and treatment scenarios, including applications such as point-of-care ultrasound. The framework emphasized the importance of transparency, adaptability, and workflow compatibility, highlighting that poorly aligned automation can disrupt team coordination and decision-making. By translating real-world observations into concrete functional requirements, this work produced a practical roadmap for incorporating automation into medical operations in future exploration missions. This project’s outcomes directly inform NASA’s efforts to develop self-sufficient medical systems that support crew autonomy while preserving effective teamwork under operational stress.

Impact: Our project, focusing on supporting team-based decision-making, demonstrated that effective performance in isolated, confined, and extreme operational contexts depends on a combination of clear role definition, adaptive leadership, shared mental models, and decision processes that balance structured authority with collaborative input. Across the Antarctic, construction, low-resource medical/emergency response, and expeditionary science teams, high-performing groups consistently exhibited rapid problem recognition, flexible redistribution of responsibilities, and open communication climates that enabled concerns to be raised without fear of negative consequences. Teams that maintained psychological safety and used simple, shared frameworks for prioritization were more resilient in the face of unexpected events, while those with ambiguous authority structures or poorly aligned procedures experienced slower responses and greater coordination breakdowns. In medical scenarios, the research further showed that automation is most effective when it functions as a collaborative partner that enhances situational awareness, supports information sharing, and preserves human authority, rather than attempting to replace human decision-making.

These findings provide direct implications for the design of future Lunar, Cis-Lunar, and Mars mission concepts. The results support the development of operational models that use consensus-driven decision making by default, with expertise-based authority overrides during time-critical situations, along with distributed role adaptability and structured debrief processes to capture lessons learned. The project translated empirical observations from terrestrial analog teams into measurable indicators of team performance and system-level recommendations for training, procedures, and decision-support technologies. These insights inform the development of pre-mission training curricula, just-in-time refresher modules, and onboard support systems that promote resilient team functioning under conditions of communication delay, resource limitation, and high autonomy. By integrating systems engineering approaches with human factors data, this project provided a framework for modeling and training team-based problem solving that can be

	directly applied to long-duration exploration missions, enabling crews to maintain performance, safety, and adaptability as operational demands evolve over the mission lifecycle.
Bibliography Type:	Description: (Last Updated: 03/18/2026)
Articles in Peer-reviewed Journals	Porter A, Arquilla K, Stankovic A. "Qualitative assessment of terrestrial care settings to inform self-sufficient spaceflight medical care." Wilderness Environ Med. 2025 Sep;36(1_suppl):61S-76S. https://doi.org/10.1177/10806032251351589 ; PubMed PMID: 40641436 , Sep-2025